

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M. Tech)

Semester – II University Examination January 2010

M. Tech. (CAD/CAM)

ME – 707 Computer Aided Manufacturing

Date: 03.01.2011, Monday

Time: 10:30 a.m. to 01:30 p.m.

Maximum Marks: 70

Instructions:

1. Figures to the right indicate **full** marks.
2. Make suitable assumptions and draw neat figures wherever if required.

SECTION-I

- Q1 (a)** Explain Computer Assisted Programming – APT stating illustration. 03
- (b)** Define Computer Aided Manufacturing. Classify its applications into broad categories & explain in brief each of them. 04
- (c)** Explain the types of format used in NC/CNC programming stating illustrations and their relative merits and demerits. 04

- Q2 (a)** Draw a sketch of vertical spindle machining center with a rotary table and specify all the axes on it. State the rules of axes designation followed. 04
- (b)** Explain with sketch the types of control with reference to motion in NC/CNC machines. 04
- (c)** Write step by step procedure for Geometrical Tolerancing. Define Datum & Datum system. 04

OR

- Q2 (a)** Describe with sketch recirculating ball screw used in CNC machines stating advantages. What is preloading? 04
- (b)** Sketch and explain (i) tool offset and (ii) cutter radius compensation cycles. 04
- (c)** Explain the benefits of Computer Aided Inspection. 04

- Q3 (a)** Explain with sketch the linear and circular interpolations stating illustrations and write the NC statements. 03
- (b)** Describe in brief the commonly used CAD/CAM integrated software packages stating their features, advantages and applications. 03
- (c)** Write a manual part program for edge milling a low-carbon steel rectangular component of size 120 x 80 x 10mm. Select and mention workpiece zero, start point and sketch tool path. 06

OR

- Q3 (a)** Explain a drilling canned cycle. Write its statement and show the motions with the help of a sketch. 03
- (b)** Explain Direct Numerical Control and Distributive Numerical Control with the help of sketches. State their advantages and limitations. 03
- (c)** Write manual part program using absolute mode for profile turning a component as shown in figure 1. Sketch tool path showing work piece zero clearly. Assume speed, feed and necessary data stating clearly. 06

SECTION – II

- Q4 (a)** What are the two hurdles in implementing GT? Explain how those are solved. 02
- (b)** What are the potential benefits of implementing CIM? 03
- (c)** Write the differences between 06
- (i) Serial and Parallel communication
 - (ii) Logical and Physical view
 - (iii) BCD and Gray code

- Q5 (a)** State the reasons for using Rapid prototyping. What is the basic methodology followed for current Rapid Prototyping techniques? 06
- (b)** Explain the relevance of the following in group technology: 03
- (i) Part family
 - (ii) Composite component
 - (iii) Cellular manufacturing
- (c)** With the help of suitable illustrative example, explain three basic structures used in classification and coding schemes for the formation of part families. 03

OR

- Q5 (a)** With neat schematic diagram explain Fused Deposition Modeling technique of rapid prototyping. State its advantages and disadvantages. 06
- (b)** The following table lists ten parts that are being considered for cellular manufacturing in a machine shop. Parts are identified by letters, and machines are identified numerically. 06

Part	Machine Routing	Part	Machine Routing
A	M1→M3→M4	F	M4→M6→M1
B	M3→M4→M6→M5	G	M3→M5
C	M6→M1→M4→M3	H	M5
D	M2	I	M4→M1→M3→M6
E	M5→M2	J	M5→M2

For the data given:

1. Develop the part-machine incidence matrix
 2. Apply ROC algorithm to the part-machine to identify the machine cells and part families
- Q6 (a)** Explain the two basic methods of preventing collision between Automated Guided Vehicles. 04
- (b)** Explain the terminology used in network communication: 04
- (i) Baud rate
 - (ii) Band width
 - (iv) Local area network
 - (v) Wide area network
- (c)** Two distinct GT cells are formed for 5 parts and 4 machines as shown below: 04

Machines	Parts				
	P2	P4	P1	P3	P5
M2	1	1			
M4	1	1			
M1			1	1	1
M3			1	1	

Calculate the Bond Energy of the matrix.

OR

- Q6 (a) What are different methods of communication with AGVs? Explain any one method. 04
- (b) What are the basic differences between Flexible Manufacturing Cell (FMC) and Flexible Manufacturing System (FMS)? 04
- (c) Explain the following flexibilities with reference to FMS: 04

Product flexibility, Product mix flexibility and Routing flexibility

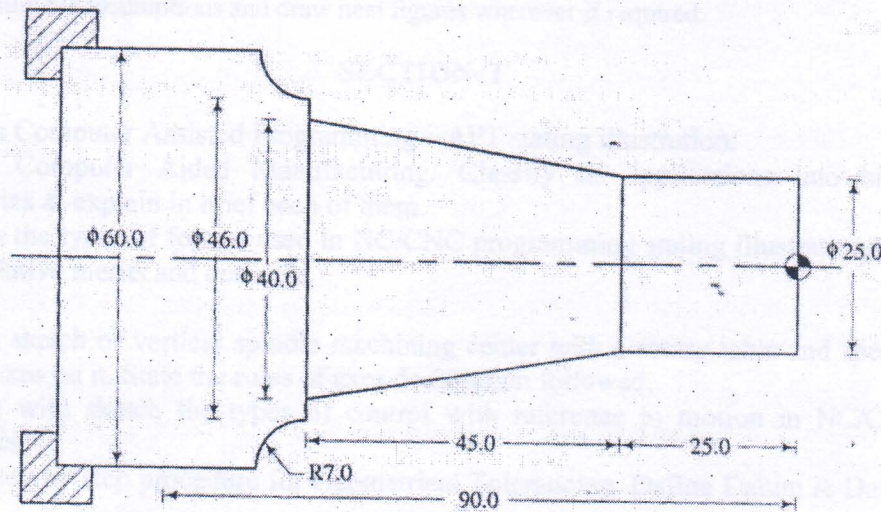


Figure 1.

G00 Rapid traverse	M00 Program stop
G01 Linear interpolation	M02 End of program
G02 Circular interpolation CW	M03 Spindle CW
G03 Circular interpolation CCW	M04 Spindle CCW
G78, G79 Mill cycle	M05 Spindle OFF
G80 Canned cycle cancel	M06 Tool change
G84 Threading cycle	M08 Coolant ON
G90 Absolute mode	M09 Coolant OFF
G91 Incremental mode	M30 End of tape
G92 Presetting	